



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 2nd Aug 2026

In addition to part I (General Handout for all courses appended to the Timetable) this portion gives further specific details regarding the course.

Course No : CHEM F337
Course Title : Green Chemistry and Catalysis
Instructor-in-Charge : PARTHA SARATHI ADDY
Instructor(s) : --
Tutorial/Practical Instructors: --

1. Course Description:

Definition and overview of the twelve principles of Green Chemistry, alternative starting materials; alternative synthesis and reagents; E-factor and the concept of atom economy; the role of catalysis, alternate energy sources (microwave & ultrasound), catalysis by solid acids and bases, bio-catalysis, catalytic reduction, catalytic oxidation, catalytic C–C bond formation, cascade catalysis, enantioselective catalysis, alternative reaction media, renewable raw materials, industrial applications of catalysis.

2. Scope and Objective of the Course:

The objectives of this course are to provide the students with a fundamental understanding of Green Chemistry with an emphasis on the design, prepare and use of chemicals and protocols that have little or no pollution potential or environmental risk. Catalysis being considered as the prime tool to efficiently inculcate the 'green' aspects, will be discussed at length including its types, guiding principles and milestone applications. The students will get exposed to the latest development trends to obtain a comprehensive understanding.

3. Text Books:

T1: Green Chemistry and Catalysis, Roger Arthur Sheldon, Isabel Arends, and Ulf Hanefeld, Wiley, 2007.
T2: Green Chemistry Theory and Practice P.T Anastas and J. C. Warner, Oxford University Press, 2000.

4. Reference Books:

R1: Catalysis - Concepts and Green Applications, G. Rothenberg, Wiley-VCH, 2008
R2: (a) Green Catalysis by Paul T Anastas, Wiley, 2010, Vol. 1 Homogeneous catalysis; (b) Green Catalysis by Paul T Anastas, Wiley, 2010, Vol. 2 Heterogeneous catalysis; (c) Green Catalysis by Paul T. Anastas, Wiley, 2010, Vol. 3 Biocatalysis

5. Course Plan:

Module	Lec. No.	Major Descriptors	Reference (Ch./ Sec.)
1	1-2	<u>Introduction and Tools of Green Chemistry</u>	T1 1.1-1.3 T2 1.1-4.12
		Definition and overview of the twelve principles of Green Chemistry, E factor and the concept of atom economy, Alternative feedstock/ starting materials, reagents, solvents, product/ target molecule and catalysts	
	Learning Outcomes: <i>(i) Describe principles of Green chemistry, (ii) Define basic terminologies associated in Green chemistry, (iii) Delineate importance of green chemistry aspects and catalysis phenomena</i>		



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	3-4	<u>Factors for Green Process</u> Evaluating effects of: chemistry, feedstock, reaction type, methods to design safer chemicals	T2 5.1-8.5
	Learning Outcomes: <i>(i) Evaluate and estimate the effects of different process parameters, (ii) Analyze the contributions of these factors to target a greener process</i>		
2	5-6	<u>Alternative modes for chemical activation</u> Alternate (Efficient/ cheaper) modes for heating/ activation of molecules in chemical reactions, Applications of microwave & ultrasound radiation in green synthesis	T2 9.1-9.5
	Learning Outcomes: <i>(i) Describe the necessity of alternate modes of chemical activation (heating) for Green processes, (ii) State how does these alternate modes of heating work, (iii) Illustrate use of microwave and ultrasound radiations in chemical reactions with respect to case studies</i>		
3	7-8	<u>Case studies: Green chemistry</u> Polysaccharide polymers, chemical from glucose, halide free synthesis of aromatic amines, alternative to Strecker synthesis, non-phosgene isocyanate synthesis, Haas Sea –Nine insecticide.	Lecture notes
	Learning Outcomes: <i>(i) Correlate Green chemistry principles to different case studies, (ii) Identify and illustrate the Green chemistry aspects in different chemical process examples</i>		
4	9-13	<u>Basics of Catalysis</u> Basic terminologies, Energetics in Catalysis, Catalysis types, Tools in Catalysis, Kinetics in Catalysis, Catalyst Stability and Deactivation - Sintering and Thermal Degradation; Catalyst Inhibition and Poisoning	R1 1.1-1.3 R1 2.1-2.4
	Learning Outcomes: <i>(i) Describe the phenomenon of catalysis in light of its characteristic aspects, (ii) Differentiate catalyst into classes, (iii) Analyze the tools used in catalysis, (iv) Describe the fundamental aspects of catalyst activity, dependence and variance with respect to parameters and time, (v) Describe the aspects of catalyst stability and various pathways for catalyst deactivation and degradation, (vi) interpretation of catalyst performance from experimental data</i>		
5	14-17	<u>Homogeneous Catalysis - Principles</u> Elementary steps in homogeneous catalysis, Structure-Activity Relationships (SAR) and their contributing factors, Asymmetric homogeneous catalysis principles	R1 3.1.1-3.1.3
	Learning Outcomes: <i>(i) Describe and differentiate between the different elementary steps involved in homogeneous catalysis, (ii) Define SAR and all factors contributing to SAR, (iii) Identifying the contributing factors for SAR in typical cases and based on the understanding of the elementary steps</i>		



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6	18-24	<u>Homogeneous Catalysis – Discussions and Case Studies</u> Major milestone processes in development of homogeneous catalysis in industry with high impacts, Catalytic cycles e.g. SHOP, Wacker oxidation (WO) process, Dupont Adiponitrile process, Ciba-Giegy Metolachlor process, Terephthalic acid process etc. Homogeneous catalysis in Hydrogenation, Hydroformylation, Carbonylation, Oxidation, C-C couplings etc.	R1 3.1.4 T1 3.3, 4.4, 5.3
	Learning Outcomes: <i>(i) Analyze the catalytic cycles in different homogeneously catalyzed reactions in light of the guiding principles, (ii) Cite appropriate case studies for various types of reactions & catalyst systems, (iii) Discuss catalytic cycle events for any of the described processes and identify challenges</i>		
	25-26	<u>Homogeneous Catalysis – Emerging trends</u> Metal-free Catalysts: Acid-base catalysts, Organocatalysts, Catalysis in Click Chemistry, Cascade, and Combinatorial Catalysis, Homogeneous Catalysis - pros and cons	R1 3.2-3.4 T1 9.1-9.2
	Learning Outcomes: <i>(i) Describe emerging trends in homogeneous catalysis, (ii) Discuss key catalytic aspects for different cases and identify challenges, (iii) Define cascade, consecutive and one-pot reactions in catalysis, (iv) Identify the advantages and disadvantages in each of these approaches</i>		
6	27-33	<u>Heterogeneous Catalysts – Principles, Synthesis and Characterizations</u> General understanding, Classification, Catalyst preparation principles, Porous matrices, Catalyst Formation, Characterization techniques employed – Surface analyses, microscopy and spectroscopy, Diffraction techniques, NMR techniques	R1 4.1, Lec. Notes
	Learning Outcomes: <i>(i) Describe the broad classification of heterogeneous catalysts, (ii) Describe the different synthesis techniques involved in design of heterogeneous catalysts, (iii) Analyze catalyst characterization results, (iv) Illustrate choice of characterization tools appropriately</i>		
	34-36	<u>Heterogeneous Catalysts – Illustrations and Case Studies</u> Catalyst systems – L-S, L-L, Phase transfer catalysis (PTC), Case studies – BP AVADA process, ABB AlkyClean process, IFP, Yellowdiesel process, UOP Smart process etc.	R1 4.2-4.3 T1 7.3-7.4
	Learning Outcomes: <i>(i) Describe the catalytic aspects for different types of L-S/ L-L systems, (ii) Discuss case studies in light of the understanding of heterogeneous catalysis</i>		
7	37-38	<u>Heterogeneous Catalysts – Emerging Trends</u> Nanomaterials as catalysts, Novel reaction media in catalysis- Supercritical fluids, Ionic liquids, Xerothermic media in catalysis	Lec. Notes T1, 7.5-7.8
	Learning Outcomes: <i>(i) Describe the emergence of nanocatalysts, (ii) Rationalize use of reaction media in catalysis, role of solvent (ii) Identify potential hazards in solvent use and understand challenges, (iii) Describe already known designs of alternative media, based on their properties, (iv) Cite advantages and disadvantages for each of them and generate logical choices and designs</i>		
7	39-40	<u>Biocatalysis: Basics</u> Common terminologies, Concepts of Active sites and intermediates, Mechanisms, Applications of enzymatic catalysis, Combining Bio- and	R1 5.1-5.5



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		Conventional catalysis, Non-enzymatic biocatalysts – Abzymes etc., Industrial case studies	
		Learning Outcomes: <i>(i) Describe the basic terminologies and concepts of biocatalysis, (ii) Illustrate conceptual understanding of active sites and common mechanisms in biocatalysis, (iii) Analyze case studies based on acquired concepts</i>	
8	41-42	Future Directions in Catalysis - Sustainability Renewable Raw Materials for Chemical Industry - Biomolecules, Chemical and Chemoenzymatic Transformations into Fine Chemicals and Chiral Building Blocks, Oleochemicals, Other platform chemicals, Renewable Raw Materials as Catalysts, Green Polymers from Renewable Raw Materials	T1 8.1-8.8
		Learning Outcomes: <i>(i) Describe the use of different renewable raw materials as alternate to fossilized C-resources, (ii) Interpret the challenges for large-scale application of renewable feedstock, (iii) Describe case studies</i>	

6. Evaluation Scheme:

Component	Duration	Weightage (%) (Marks)	Date & Time	Nature of component (Close Book/ Open Book)
Mid-Semester Test	90 min	30	As per the time table	Open &/or Close Book
Comprehensive Examination	180 min	40	As per the time table	Open &/or Close Book
Continuous Evaluation*&***	-	30	Will be announced through Nalanda	Close Book

* In continuous evaluation, four quizzes will be conducted in entire semester. Best THREE will be considered for final evaluation this for fourth year or onward students.

**In continuous evaluation one quiz will be linked to attendance, for the students whose ID No. 2024B2XXXX or 2025B2XXXX. For them out of four best two will be considered and third one will be considered only if the student maintains a minimum of 50% attendance throughout the semester.

7. Minimum criterion for awarding valid grades: Where a student continued to remain registered in a course but gave the instructor inadequate opportunity to evaluate him/her by absenting himself/herself from quizzes/ tests/ examinations/other components of evaluation, or by appearing in the same for the sake of appearance without applying himself/herself to the task in hand or by submitting a blank script (answer book) this event will be reported as NC (Not Cleared). Moreover, a student should obtain 30% of the average of the top 10 performers in the class, or 40% of the median marks of the class, whichever is lower to clear the course. If any student gets the marks lower than the prescribed standard mentioned above, he/she may be awarded NC.



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8. Chamber Consultation Hour: Wednesday 10 am to 11 am.

9. Notices:

All notices regarding the course will be displayed in Nalanda and/ or Department of Chemistry Notice board.

10. Make-up Policy:

No makeup for quizzes for any reason as one buffer is there.

Make up for midsem and compre would be considered only for genuine reasons decided by the instructor.

Partha Sarathi Addy
Instructor-in-charge
Course No. CHEM F337